

Marginal Structural Models

Jeffre MM et al AS 2004; 58:272

Robins JM et al EPID 2000; 11: 550

Example 1

Some Notations

- A : Exposure/treatment level
- Y : Outcome of interest measured
- L : Confounder, common cause of A and Y

Assume no unmeasured confounders given L

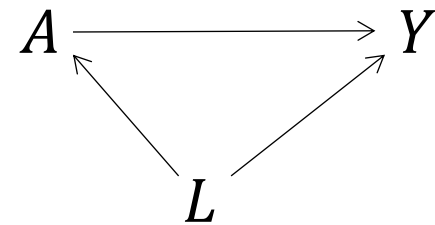


Figure 1.

Question: How to estimate the causal effect of A on Y ?

Solution 1 (Standardization method):

$$\begin{aligned} E(Y^{a=1}) - E(Y^{a=0}) \\ &= \sum_l pr(Y = 1|A = 1, L = l)pr(L = l) \\ &\quad - \sum_l pr(Y = 1|A = 0, L = l)pr(L = l) \end{aligned}$$

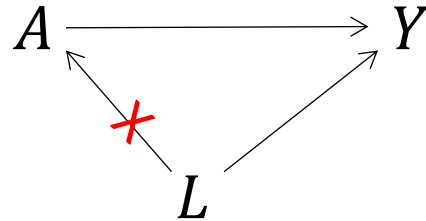
i.e. L -standardized risk in the **treated** minus that in the **untreated**,
that is,

weighted average of the L -stratum-specific risks in the treated *minus*
weighted average of the L -stratum-specific risks in the untreated,
with weights *the population share in the stratum l*

Equivalently,

Estimate the association of A with Y within each stratum l

Solution 2 (MSM method):



Two steps:

1. Remove the association between A and L
2. Estimate the association A with Y without conditioning on L

How to remove the A-L association?

This is exactly what marginal structural models accomplish!

Example 1 (cont.):

Table below summarizes observed data of (Y, A_0, L_0) . Assume the three variables follow the relationship of Figure 1. Estimate the causal effect of A on Y using **MSM** method and **Standardized** method.

	$L_0 = 1$		$L_0 = 0$	
	$A_0 = 1$	$A_0 = 0$	$A_0 = 1$	$A_0 = 0$
$Y = 1$	108	24	20	40
$Y = 0$	252	16	30	10
Total	360	40	50	50

Standardization Method (stratum-specific analysis):

$$\begin{aligned} & E(Y^{A_0=1}) - E(Y^{A_0=0}) \\ &= \sum_l pr(Y = 1 | \mathbf{A}_0 = \mathbf{1}, L_0 = l) pr(L_0 = l) \\ &\quad - \sum_l pr(Y = 1 | \mathbf{A}_0 = \mathbf{0}, L_0 = l) pr(L_0 = l) \\ &= \left(\frac{20}{50} \times \frac{100}{500} + \frac{108}{360} \times \frac{400}{500} \right) - \left(\frac{40}{50} \times \frac{100}{500} + \frac{24}{40} \times \frac{400}{500} \right) \\ &\quad = .32 - .64 = -.32 \end{aligned}$$

How about MSM method?

Example 1 (Cont.): MSM method

Step 1: Model $\text{logit}(Pr(\mathbf{A} = \mathbf{1}|L = l)) = \alpha + \beta l$

Step 2: Derive weights $w_i = \frac{1}{pr(A = a|L = l)}$ for units with $(A, L) = (0,0), (0,1), (1,0), (1,1)$, respectively,

	$L_0 = 1$		$L_0 = 0$	
	$A_0 = 1$	$A_0 = 0$	$A_0 = 1$	$A_0 = 0$
Y = 1	108	24	20	40
Y = 0	252	16	30	10
Total	360	40	50	50

Step 3: Create pseudo-population

L_0	A_0	Y	N Observed Population	$pr(A_0 L_0)$	w	N Pseudo Population
1	1	1	108	0.9	1.11	120
1	1	0	252	0.9	1.11	280
1	0	1	24	0.1	10	240
1	0	0	16	0.1	10	160
0	1	1	20	0.5	2	40
0	1	0	30	0.5	2	60
0	0	1	40	0.5	2	80
0	0	0	10	0.5	2	20

Step 4: Estimate model parameter in MSM in the pseudo-population

$$E(Y^a) = pr(Y = 1|A_0 = a)$$

$$\tau = E(Y^{a=1}) - E(Y^{a=0})$$

$$= pr(Y = 1|A_0 = 1) - pr(Y = 1|A_0 = 0)$$

$$= \frac{120 + 40}{120 + 280 + 40 + 60} - \frac{240 + 80}{240 + 160 + 80 + 20} = -0.32$$

~ Same results as obtained by standardization method?

Above – risk difference (RD). How to obtain causal risk ratio (RR), and causal odds ratio (OR) in the population?

A little more complicated example using MSM

More notations:

- $\mathbf{X}$: Measured pretreatment covariates
- $V \in \mathbf{X}$: a subset of $\mathbf{X}$, including pretreatment covariates whose association with Y is of interest
- Y^a : Potential outcome if receive level a
- $f(Y^a|V)$: Marginal density of a potential outcome in the subset defined by V .

Estimate causal effect $\leftrightarrow$ Compare $E(Y^a|V)$ vs. $E(Y^{a'}|V)$

Marginal density $E(Y^a|V) \sim \text{Identifiable}$ under assumptions:

1. **Conditional Exchangeability:** Potential outcomes do not predict treatment assignment once one accounts for baseline covariates X

$$Y^a \perp A|X \text{ for all } a$$

(Rosenbaum and Rubin, 1983)

2. **Positivity:** There is nonzero probabilities that subjects with any covariates level X receive treatment level a , i.e.
 $1 > \Pr(A = a|X) > 0$.

Marginal Structural Model (MSM)

- ~ Model the **marginal** expectations of the potential outcomes Y^a given specified baseline covariates V :

$$E(Y^a|V) = q(a, V)$$

For example, $E(Y^a|V) = q(a, V) = \text{expit}(a\beta_a + V\beta_v)$,

where

β_a : Effect of treatment group $A=a$ on odds of success

β_v : Effect of covariates V on odds of success

$$\text{expit}(r) \equiv \frac{\exp(r)}{1+\exp(r)} \text{ such that } \text{logit}(\text{expit}(r)) = r$$

- ~ They are **structural** models, because they model the probabilities of potential outcomes Y^a

Estimation under MSM (4-step procedure)

Step 1: Model the association of covariates and treatment assignment $Pr(A = a|X)$

Step 2: Use the specified model to derive weights

Step 3: Use the weights to create an artificial (pseudo-) population in which covariates are not associated with the treatment indicator $A \perp X$.

Step 4: Estimate the parameters in MSM using artificial (pseudo-) population

Example 2: *A study of Arteriovenous Fistula Surgery*

Data (Feldman et al. 2003) of $n=348$ subjects undergoing AVF surgery

Background: subjects with renal disease require a kidney transplant to replace the lack of function of their failed native kidneys. *Hemodialysis* is the most common form of kidney replacement, and its successful implementation depends on the ability to maintain a surgically created vascular conduit capable of sustaining high blood flows. *Native arteriovenous fistula* (AVF) is one such conduit. *Heparin* is used to prevent blood clots from forming during or after surgery, which can affect the successful maturation of the graft; potential benefit of *Heparin with different levels of dose* were not, however, confirmed at the time of the study.

Aim: Identify patients whose AVF surgeries are likely to succeed and so perform the surgery only for the subgroup.

Y (=1 if surgery is successful - successful use of AVF in hemodialysis; =0 otherwise)

A: heparin use (=1 if at least 1,000 units, high dose; =0 otherwise)

X: personal characteristics

- Age (<=40; 41-56; 57-69; 70+ as reference)
- Prior permanent access (Yes or No)
- Dependent on dialysis (Yes or No)
- Cardiovascular disease (Yes or No)
- Blood Pressure at admission (Low or normal), and Size of the vessels, collected in the surgical procedure but not measured before surgery

Aim: Identify patients whose AVF surgeries are likely to succeed and so perform the surgery only for the subgroup

That is,

Identify patients with large $E(Y^a = 1|V)$ if treated by treatment level a that is deemed to be preferable,

Equivalently,

Estimate the probability of success of the procedure if someone were provided the optimal heparin dose.

Estimation (4 steps)

Step 1: Construct the Model: $\text{logit}\{\Pr(A = 1|X)\} = X\beta$

Step 2: Compute weight for unit i by the inverse of estimated probability of treatment assignment

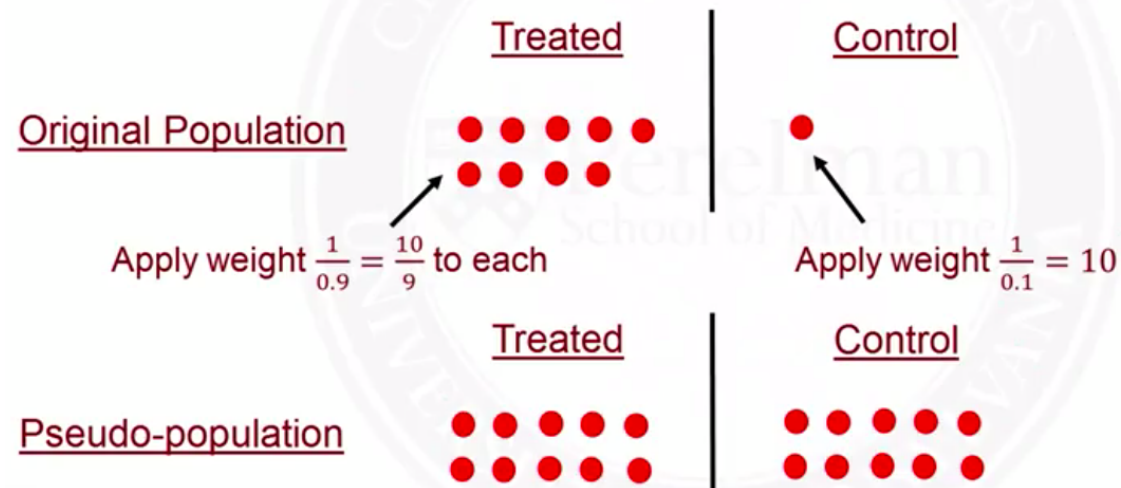
$$w_i = \frac{I\{A_i = 1\}}{p_i} + \frac{1 - I\{A_i = 1\}}{1 - p_i},$$

where probability of treatment $p_i = \Pr(A_i = 1|X_i)$ is estimated by $\hat{p}_i = \text{expit}(X_i\hat{\beta})$

Step 3: Create pseudo-population

Pseudo-population

- ♦ Suppose $P(A=1|X)=0.9$



- Suppose a subject's probability, given covariates, of receiving treatment is .9
- Weight: inverse of the probability that a subject with X received his/her observed treatment
- Apply the weights to the study subjects (original population) creates an artificial population which perfectly balanced X used to derive the weights

In the pseudo-population, everyone is equally likely to be treated, no matter X values

Table 2. Logistic Regression Model for Associations of Various Predictors With High-Dose Heparin Use

	Coefficient ($\hat{\beta}$)	Standard error	Odds ratio	p value
Intercept	.87	.52		
Age (reference: 70+)				<.0001
≤40	−1.72	.34	.18	
41–56	−1.54	.48	.21	
57–69	−.68	.38	.51	
Prior permanent access?				.01
Yes	.67	.27	1.94	
Upstream vein diameter (reference: < 3 mm)				.04
3–4	−.46	.59	.63	
4–5	−.60	.64	.55	
>5	−1.27	.67	.28	
Dependent on dialysis?				.23
Yes	−.42	.35	.66	
Cardiovascular disease?				.57
Yes	−.10	.18	.90	
Blood pressure at admission				.73
Low (versus normal)	−.13	.38	.87	

- $\text{logit}\{\text{Pr}(A = 1|X)\} = X\beta$
- Older people receives high dose heparin use
- Smaller vassals receive high dose heparin use

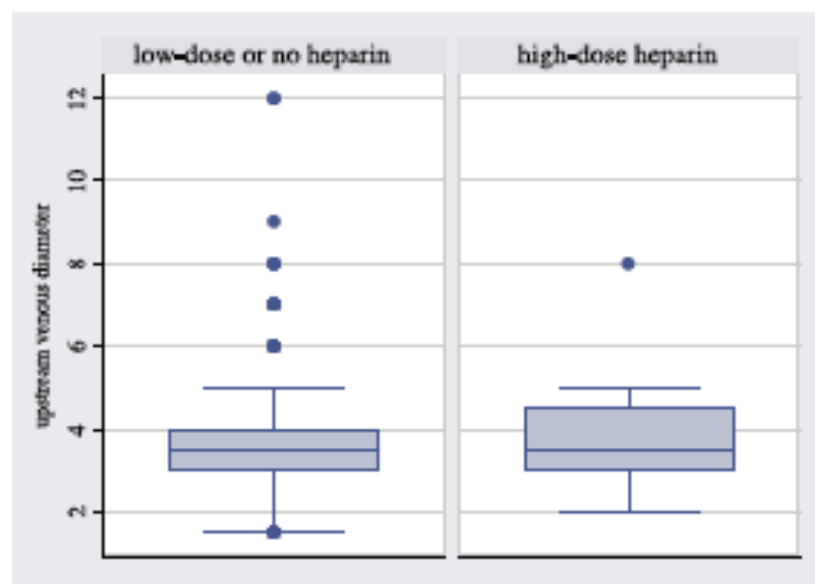
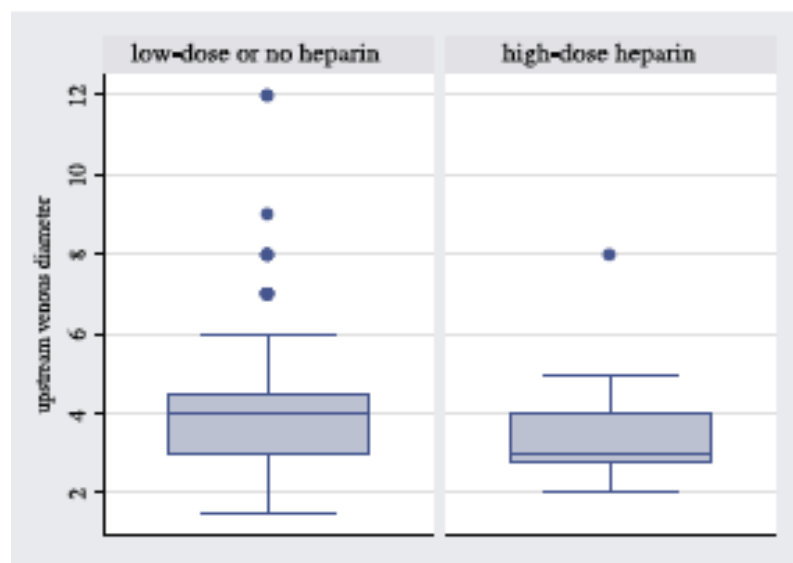
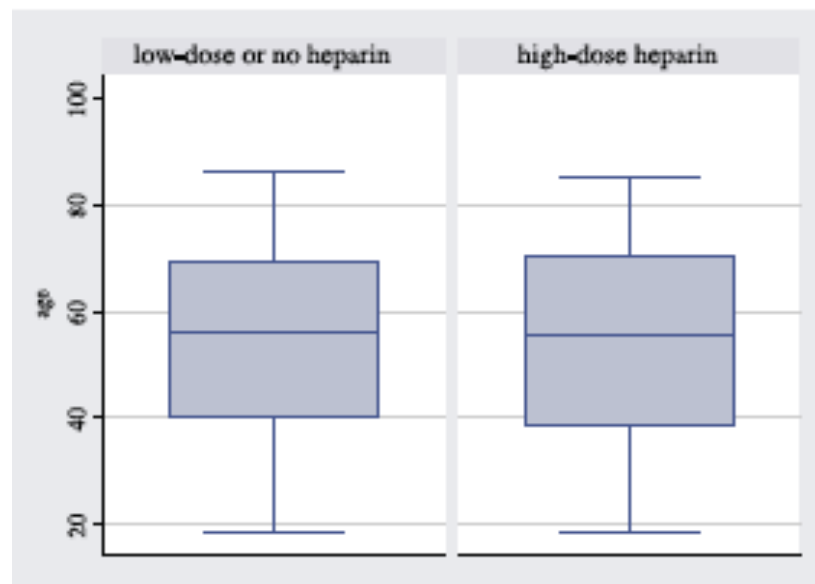
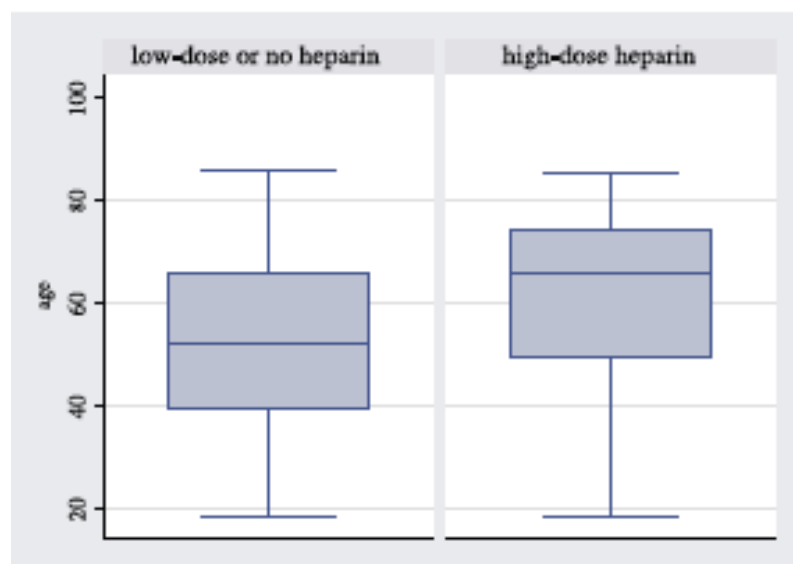


Figure 1. Boxplots of distributions of continuous covariates by heparin dose.

Figure 2. Weighted boxplots of distributions of continuous covariates by heparin dose.

- Figure 1 shows that subjects receiving high-dose heparin are more likely to be older and have smaller vessels used in the surgery;

Because older age and smaller vessels are associated with poorer outcomes, one would expect analyses that do not control for these variables to understate the benefit of heparin.

- Figure 2 shows that imbalance of age and vessel size between treatment groups is removed after weighting.

Now let us estimate causal effect of heparin use (A) on the outcome (Y)

Step 4: Estimate model parameter in MSM

Under assumptions:

- Conditional Exchangeability: Given X the treatment assignment A is random; X fully captures the confounding $Y^a \perp A|X$
- Positivity: propensity scores are strictly positive $(0, 1)$

Under MSM:

$$E(Y_i^a|V) = \mu_i(\beta) = \text{expit}(a\beta_a + V\beta_v) \text{ for } a=0, 1$$

Solving estimating equations below for β in the pseudo-population

$$\sum_{i=1}^n \mathbf{w}_i \frac{\partial \mu_i^T}{\partial \beta} \text{Var}^{-1}(Y_i) \{Y_i - \mu_i(\beta)\} = 0$$

$$\Leftrightarrow \sum_{i=1}^n w_i \{Y_i - \mu_i(\beta)\} (A_i, V_i)^T = 0$$

The estimates are denoted by $\hat{\beta}$.

Causal OR:

$$\text{OR} = \frac{p(Y_i^1 = 1|V)}{p(Y_i^1 = 0|V)} \bigg/ \frac{p(Y_i^0 = 1|V)}{p(Y_i^0 = 0|V)},$$

estimated by

$$\widehat{\text{OR}} = \frac{\frac{\exp(\widehat{\beta}_1 + V\widehat{\beta}_v)}{1 + \exp(\widehat{\beta}_1 + V\widehat{\beta}_v)}}{\frac{1}{1 + \exp(\widehat{\beta}_1 + V\widehat{\beta}_v)}} \bigg/ \frac{\frac{\exp(V\widehat{\beta}_v)}{1 + \exp(V\widehat{\beta}_v)}}{\frac{1}{1 + \exp(V\widehat{\beta}_v)}} = \exp(\widehat{\beta}_1)$$

The resulting $\exp(\widehat{\beta}_1)$ estimates causal effect of treatment A=1 on the outcome in OR scale.

Variance of $\widehat{\beta}_1$ can be estimated by the robust variance estimator (Liang and Zeger, 1986)

Table 3. Marginal Structural Logistic Regression Model for Successful AVF Maturation

	Coefficient ($\hat{\beta}$)	Standard error	Odds ratio	p value
Intercept	.26	.45		
High-dose heparin Yes	1.05	.17	2.86	<.0001
Age (reference: 70+)				.075
≤40	1.15	.52	3.15	
41–56	.63	.40	1.87	
57–69	.10	.25	1.11	
Prior permanent access? Yes	.48	.23	1.61	.035
Dependent on dialysis? Yes	−.68	.31	.51	.028
Cardiovascular disease? Yes	−1.24	.38	.29	.001
Blood pressure at admission Low (versus normal)	−.45	.38	.64	.250

- High-dose heparin increases the odds of successful maturation by 2.86 times
- Younger age, not being on dialysis, not having cardiovascular disease, having had previous permanent access are all associated with higher prob. of success.

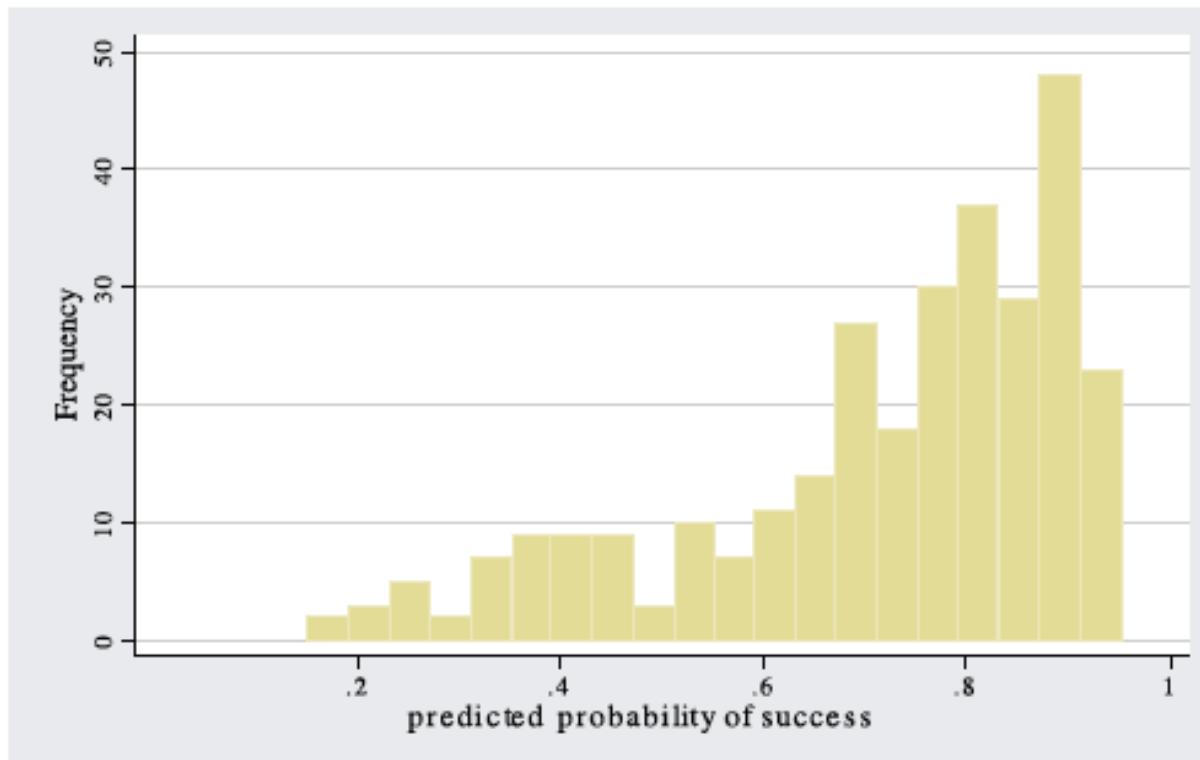


Figure 3. Histogram of predicted probabilities of success, $E(Y^1|X; \hat{\beta})$ in AVF study.

Figure 3 shows the distribution of predicted Y under high-dose A=1

- Substantial variation among subjects
- The model distinguishes subjects with high success from those less likely to succeed

Table 4. Odds Ratios From Various Weighted and Unweighted Logistic Regression Models for the Association of Heparin With Surgical Success

Covariates in structural model (V)	Standard models (no weights)			Marginal structural models (weights)		
	Point estimate	95% confidence interval		Point estimate	95% confidence interval	
1. None	1.7	1.3	2.2	2.4	1.7	3.4
2. Age (3 terms), dialysis dependent, history of cardiovascular disease, prior vascular access, admission mean arterial pressure	2.2	1.6	3.0	2.9	2.1	4.0
3. Variables in 2, upstream venous diameter (three terms)	2.7	2.1	3.6	2.9	2.1	4.1

NOTE: Heparin use is classified as high (at least 1,000 units during surgery) or low (less than 1,000 units; includes no use).

Three MSMs include models controlling for

1. No covariates in V , but A only
2. The same covariates V as in our logistic MSM
3. All covariates X , including the postsurgery variable

Observations by comparing weighted and unweighted analyses

- Weighted form of all models provides a valid estimate of causal odds ratio
- Model 2 conditional on presurgery covariates V is useful for identifying subjects whose surgery is likely to succeed
- Under Model 2 we have $OR(A)=2.2$ vs. 2.9: the unweighted model $E(Y|V)$ doesn't control for confounding by vessel size; in contrast, our MSM estimates control confounding by vessel size through weighting
- Under Model 3 we have similar effects $OR(A)=2.7$ vs. 2.9: both models control for confounding by vessel size. However, the model does not make the predictions on $E(Y^a|V)$, which is of greatest clinical interest.

Time dependent treatments

More notations:

- A_1 : Exposure/treatment level at time 1 (e.g. anti-HIV drugs)
- A_2 : Exposure/treatment level at time 2
- Z : a covariate measured between times 1 and 2 (e.g. HIV molecules detected in serum)
- U : an unmeasured common cause of Z and Y (e.g. immune function)
- Y : outcome measure after time 2
e.g. CD4 cell count/ $(m^3 \times 100)$, so 2.5 is 250 cells/ m^3

Goal: Estimate the causal effect of time-varying treatment (both A_1 and A_2) on the outcome Y

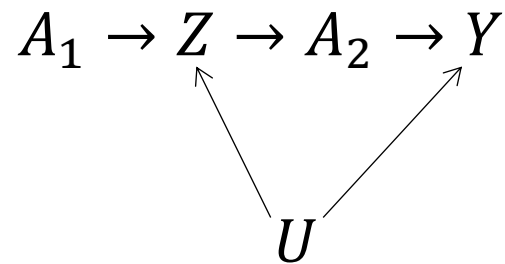


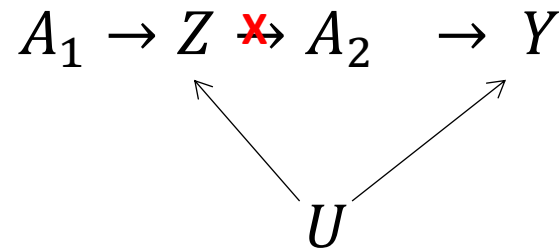
Figure 2

Dilemma:

- If we estimate the association of (A_1, A_2) with Y without conditioning on Z , then there is an open pathway through A_2 - Z - U - Y
- If we estimate the association of (A_1, A_2) with Y with conditioning on Z , then there is an open pathway through A_1 - U - Y

What can we do?

A SOLUTION (MSM):



Remove Z - A_2 association $\rightarrow$ we estimate the causal effect from (A_1, A_2) on Y without conditioning on Z , since A_2 - Z - U - Y pathway is blocked by removing Z - A_2 association

How to remove the A_2 - Z association?

This is exactly what marginal structural models accomplish!

Estimation:

Observed Data

N	Z	A1	A2	g(A)	Y
25	0	0	0	0	1.55
8	0	0	1	.5	2.55
4	1	0	0	0	2.85
13	1	0	1	.5	3.85
13	0	1	0	.5	1.65
4	0	1	1	1	2.65
8	1	1	0	.5	2.95
25	1	1	1	1	3.95

$g(A)$: a chosen function of A_1 and A_2 ,

For example, $g(A) = (A_1 + A_2)/2$

Define $A = \{A_1, A_2\}$, estimate causal effect of $g(A)$ on Y !

Association Models with Parameter α 's

– Without conditioning on Z:

Model 1: $E(Y|g(A)) = \alpha_0 + \alpha_1 g(A)$

```
> lm(Y~g.A, weight=N)
```

Call:

```
lm(formula = Y ~ g.A, weights = N)
```

Coefficients:

(Intercept)	g.A
1.729	2.041

→ There is about 204 cells/ m^3 difference

– Conditioning on Z:

Model 2: $E(Y|g(A), Z) = \alpha_0 + \alpha_1 g(A) + \alpha_2 Z$

```
> lm(Y~g.A+Z, weight=N)
```

Call:

```
lm(formula = Y ~ g.A + Z, weights = N)
```

Coefficients:

(Intercept)	g.A	Z
1.532	1.006	1.429

→ There is about 101 cells/ m^3 difference

MSM - Remove association between Z and A2:

Step 1: propensity model ($p_1 = pr(A_1 = a|\emptyset)$; $p_2 = pr(A_2 = a|Z = z)$)

$$pr(A_1 = 0|\emptyset) = \frac{50}{100}; pr(A_1 = 1|\emptyset) = \frac{50}{100};$$

$$pr(A_2 = 0|Z = 0) = \frac{38}{50}; pr(A_2 = 1|Z = 0) = \frac{12}{50};$$

$$pr(A_2 = 0|Z = 1) = \frac{12}{50}; pr(A_2 = 1|Z = 1) = \frac{38}{50};$$

Step 2: derive weights: ($w = 1/(p_1 p_2)$)

N	Z	A1	A2	g(A)	Y	w
25	0	0	0	0	1.55	2.63
8	0	0	1	.5	2.55	8.33
4	1	0	0	0	2.85	8.33
13	1	0	1	.5	3.85	2.63
13	0	1	0	.5	1.65	2.63
4	0	1	1	1	2.65	8.33
8	1	1	0	.5	2.95	8.33
25	1	1	1	1	3.95	2.63

Step 3: create pseudo-population

Step 4: estimate MSM parameters without conditioning on Z

$$E(Y^a) = \beta_0 + \beta_1 g(a)$$

```
> lm(Y~g.A, weight=N*w)
```

Call:

```
lm(formula = Y ~ g.A, weights = N * w)
```

Coefficients:

(Intercept)	g.A
1.987	1.526

→ There is about 153 cells/ m^3 difference

In Summary: MSMs can be used to estimate causal effect of a time-varying treatment on an outcome

Using Z to calculate the weights in MSM, rather than by adding Z to the regression model as regressors

Check association between A2 and Z:

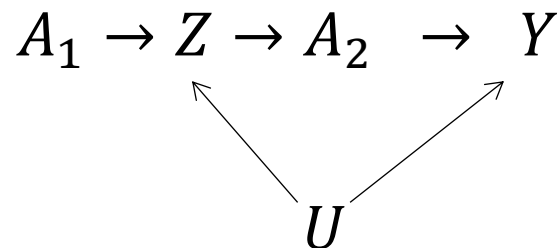
- In unweighted data:

```
> asso=glm(AZ~Z, weight=N, family=binomial(link="logit"))
> exp(cbind("Odds ratio" = coef(asso), confint.default(asso, level = 0.95)))
```

	Odds ratio	2.5 %	97.5 %
(Intercept)	0.3157895	0.1650226	0.604299
Z	10.0277777	4.0050011	25.107690

Here, N is the count in each cell defined by A_1 , A_2 , and Z

Note that the value of one is not in the estimated 95% CI (4, 25.1).

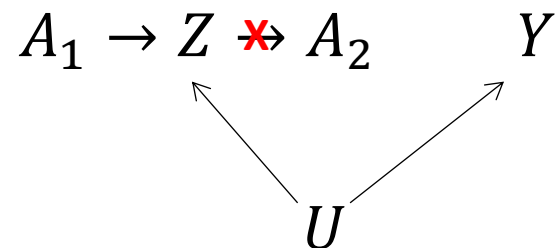


- In weighted data

```
> asso.w=glm(A2~Z, weight=N*w, family=binomial(link="logit"))
Warning message:
In eval(family$initialize) : non-integer #successes in a binomial glm!
> exp(cbind("Odds ratio.w" = coef(asso.w), confint.default(asso.w, level = 0.95)))
```

	Odds ratio.w	2.5 %	97.5 %
(Intercept)	1.0002001	0.7580166	1.31976
Z	0.9995999	0.6753724	1.47948

Note that the value of one is in the estimated 95% CI (.675, 1.479)



Stabilized weights

- The probabilities $pr(A = a|Z = z)$ may vary greatly between subjects → Extremely large values of the weight → Large variance of β estimates
- Stabilized weights

$$w_i^s = pr(A = a_i)/pr(A = a_i|Z = z_i)$$

Revisit Example 3 – MSM estimation

Step 1: propensity model ($p_1 = pr(A_1 = a|\emptyset)$; $p_2 = pr(A_2 = a|Z = z)$)

$$pr(A_1 = 0|\emptyset) = \frac{50}{100}; pr(A_1 = 1|\emptyset) = \frac{50}{100};$$

$$pr(A_2 = 0|Z = 0) = \frac{38}{50}; pr(A_2 = 1|Z = 0) = \frac{12}{50};$$

$$pr(A_2 = 0|Z = 1) = \frac{12}{50}; pr(A_2 = 1|Z = 1) = \frac{38}{50};$$

Step 2: derive stabilized weights: $\left(sw = \frac{pr(A_1=a)pr(A_2=a)}{p_1p_2} \right)$

N	Z	A1	A2	g(A)	Y	w	sw
25	0	0	0	0	1.55	2.63	0.66
8	0	0	1	.5	2.55	8.33	2.08
4	1	0	0	0	2.85	8.33	2.08
13	1	0	1	.5	3.85	2.63	0.66
13	0	1	0	.5	1.65	2.63	0.66
4	0	1	1	1	2.65	8.33	2.08
8	1	1	0	.5	2.95	8.33	2.08
25	1	1	1	1	3.95	2.63	0.66

MSM vs. Propensity Score Weighting Method

Similarities:

- Both methods involve creating weights derived from estimated propensity scores to balance covariates
- In special case when treatment is **dichotomous and time-independent**, MSM estimator becomes essentially **equivalent** to PS weighting estimator

Differences:

- MSM explicitly models potential outcomes, making them more flexible and suitable for generalization to studies with non-dichotomous or time-dependent treatments.

R codes

```
#-----  
#Codes for MSM method  
#-----  
#-input data  
  N=c(25,8,4,13,13, 4,8,25)  
  Z=c(0,0,1,1,0,0,1,1)  
  A1=c(0,0,0,0,0,1,1,1)  
  A2=c(0,1,0,1,0,1,0,1)  
  g.A=(A1+A2)/2  
  Y=c(1.55,2.55,2.85,3.85,1.65,2.65,2.95,3.95)  
  
#-input weights and stabilized weights for MSM estimation  
  w=c(2.63,8.33,8.33,2.63,2.63,8.33,8.33,2.63)  
  sw=c(.6579,2.0833,2.0833,.6579,.6579,2.0833,2.0833,.6579)  
  
#-analyses under association models 1&2 and the MSM  
  asso.1=lm(Y~g.A, weight=N)  
  asso.2=lm(Y~g.A+Z, weight=N)  
  msm.w=lm(Y~g.A, weight=N*w)  
  msm.sw=lm(Y~g.A, weight=N*sw)  
  
#-association between A2 and Z in unweighted vs. weighted data  
  A2.Z=glm(A2~Z, weight=N, family=binomial(link="logit"))  
  exp(cbind("Odds ratio" = coef(A2.Z), confint.default(A2.Z, level =  
    0.95)))  
  
  A2.Z.w=glm(A2~Z, weight=N*w, family=binomial(link="logit"))  
  exp(cbind("Odds ratio.w" = coef(A2.Z.w), confint.default(A2.Z.w,  
    level = 0.95)))
```

